

Migration Strategy Proposal

1. Migration Approaches

- Approach A: “Lift and Shift”(1-to-1 copy)
 - Pros: Straightforward and easy to plan.
 - Cons: Very slow for 100 pipelines and does not use Databricks optimizations.
- Approach B: Complete Redesign
 - Pros: Gives the absolute best Databricks performance.
 - Cons: Way too expensive and takes too long to rewrite 100 pipelines from scratch.
- Approach C: Pattern-Based Templates
 - Pros: Highly scalable. We write one code template and reuse it for multiple similar pipelines.
 - Cons: Requires more complex planning upfront.

2. Recommendation Strategy

- I recommend approach C(Templates).
- For 100 pipelines, we can't rewrite them one by one. Grouping them by patterns and reusing Databricks templates is the fastest and most cost-effective way to migrate.

3. Effort Estimation

- Total Time: 3 to 5 months for 100 pipelines.
- Team: 3-4 Data Engineers
- Phases: Assessment(2 weeks), Template Building(4-6 weeks), Migration Execution(4 weeks), Testing & Cutover(2-3 weeks)

4. Feasibility, Assumptions, and Risks

- Feasibility: High, both tools use Azure, making the technical move smooth.
- Assumptions: The underlying raw data storage(Azure Data Lake) will remain in the exact same place.

- Risk: Data mismatch between the old system and the new system.
- Mitigation: We will run both Databricks and Synapse side-by-side for 2 weeks to compare the final numbers before turning Synapse off entirely.